

Using Bit Manipulation

Problem

Given a string containing lowercase English letters, find and print all the characters that occur more than once.

You must use bit manipulation to track the characters that have already appeared.

Input Format

The first line contains a string S

Constraints

$1 \leq |S| \leq 10^5$ S contains only lowercase English letters (a-z)

Output Format

Print all duplicate characters in the order of their first repeated occurrence. If there are no duplicates, print No duplicates.

Sample Input 0

programming

Sample Output 0

r g m

Sample Input 1

greater

Sample Output 1

r e

Change Theme

Language Java 15

1import java.util.*;

2

3class Solution {

4

5 public static void main(String[] args) {

6 Scanner sc = new Scanner(System.in);

7 String s = sc.nextLine();

8 int seen = 0;

9 int duplicate = 0;

10 for (int i = 0; i < s.length(); i++) {

11 int bit = s.charAt(i) - 'a';

12 if ((seen & (1 << bit)) != 0) {

13 duplicate |= (1 << bit);

14 } else {

15 seen |= (1 << bit);

16 }

17 }

18 int printed = 0;

19 boolean found = false;

20 for (int i = 0; i < s.length(); i++) {

21 char ch = s.charAt(i);

22 int bit = ch - 'a';

23 if ((duplicate & (1 << bit)) != 0

24 (printed & (1 << bit)) == 0) {

25 System.out.print(ch + " ");

Line: 29 Col: 10

Upload Code as File

Run Code

Submit Code

☐ Test against custom input

Congratulations

You solved this challenge. Would you like to challenge your friends?

Test case 0

Test case 1

Compiler Message

Success